Mahatma Gandhi University, Kottayam

B.Sc. (Hons.) Physics (Third Semester) Practical Model Question Paper

Discipline Specific Elective (DSE)

MG3DSEPHY203 – Introduction to Optics

Time: 2 Hours **Max. Marks :35**

1) Determine the focal length of the given convex lens by measuring the object distance and image distance for various object positions. Also study the nature, size, and orientation of the image formed for different object distances. [U][7]
2) Determine the focal length of the given concave lens by measuring the object distance and image distance. Also study the nature of the image for different object distances. [U][7]
3) Using a collimated light beam, such as a laser, and a plane mirror, verify that the angle of incidence is equal to the angle of reflection. Comment on any observed deviations from the expected result. [U][7]
4) Study the variation of the angle of refraction with respect to the angle of incidence for a laser beam passing from air into water contained in a small rectangular glass vessel. Use the data to estimate the refractive index of water. [U][7]
5) Using a spectrometer and a grating/prism, determine the wavelengths of any three spectral lines of the given mercury lamp. Calculate the difference between the measured value and actual value. [U][7]
6) Using a spectrometer and a grating/prism, estimate the frequency of any three spectral lines of the given mercury lamp. Calculate the difference between the measured value and actual value. [U][7]
7) Study the interference pattern formed on a screen/detector by a laser beam passing through a narrow double slit of known slit width and hence estimate the wavelength of the given laser beam. [U][7]
8) Study the interference pattern formed on a screen/detector by a laser beam passing through a narrow double slit and hence estimate the slit width of the given double slit. Wavelength of the laser beam is given laser. [U][7]
9) Study the total internal reflection of light within a transparent solid medium and estimate the critical angle within the medium. [U][7]
10) Observe and quantify the spherical aberration of a biconvex lens by measuring the focal lengths for the paraxial and marginal rays. [U][7]
11) Observe and quantify the chromatic aberration of a given biconvex lens by determining its focal length for two different wavelengths in the visible region. [U][7]
12) Measure the angular size of an object using a telescope and hence estimate its actual linear size by measuring the distance between the telescope and the object. [U][7]

13) Determination of focal length of a telescope objective lens. [U][7]

14) Determine the diameter/length of a given small object using a microscope. [U][7]

15) Using a travelling microscope, observe the concave meniscus formed by water in a narrow glass tube and measure the vertical distance between the upper edge and lowest point of the meniscus. [A][7]

16) Determine the thickness of a given small object such as a glass plate/plastic sheet/paper using a microscope. [U][7]

17) Using a travelling microscope, observe a graduated stainless-steel ruler and verify the distance between two consecutive millimetre (mm) markings. Repeat the measurement at another two different positions along the scale and comment on the consistency and accuracy of the markings. [A][7]